```
import pandas as pd
df=pd.read_csv("/content/Advanced_pandas_sales_dataset.csv")
df.head()
```

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
0	1001	2024-10-01	Priya	Hyderabad	East	Mouse	Accessories	1735	7	0.00	307.52
1	1002	2024-05-03	Sneha	Kolkata	East	Monitor	Electronics	807	2	0.15	49.89
2	1003	2024-01-26	Anita	Mumbai	West	Phone	Electronics	2328	1	0.20	66.47
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
4	1005	2024-10-15	Rahul	Bangalore	South	Monitor	Electronics	3550	2	0.20	146.30

Next steps: Generate code with df | New interactive sheet

Q1 Dataset Exploration Perform the following exploratory steps: Display the first 10 rows Display the dataset structure using info() Find the number of rows and columns Identify the data types of each column

```
# Display first 10 rows
print(df.head(10))
```

```
   Order_ID  Order_Date Customer       City Region   Product     Category  \
0      1001  2024-10-01    Priya  Hyderabad   East     Mouse  Accessories
1      1002  2024-05-03    Sneha    Kolkata   East   Monitor  Electronics
2      1003  2024-01-26    Anita     Mumbai   West     Phone  Electronics
3      1004  2024-06-06     Sara       Pune   East  Keyboard  Accessories
4      1005  2024-10-15    Rahul  Bangalore  South   Monitor  Electronics
5      1006  2024-07-04    Pooja  Bangalore   West   Printer       Office
6      1007  2024-06-24      Dev    Chennai   East   Monitor  Electronics
7      1008  2024-03-18    Kavya    Kolkata   East     Phone  Electronics
8      1009  2024-10-31    Kavya  Ahmedabad   East     Phone  Electronics
9      1010  2024-12-25    Pooja  Ahmedabad  North     Phone  Electronics

   Sales  Quantity  Discount  Profit
0   1735         7      0.00  307.52
1    807         2      0.15   49.89
2   2328         1      0.20   66.47
3   3933         3      0.20  -71.66
4   3550         2      0.20  146.30
5   3073         8      0.20  191.58
6   1170         5      0.20   22.83
7   4693         7      0.05  918.49
8   5071         6      0.10  707.55
9   2711         8      0.00  477.32
```

```
# Display dataset structure
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 230 entries, 0 to 229
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order_ID    230 non-null    int64
 1   Order_Date  230 non-null    object
 2   Customer    230 non-null    object
 3   City        230 non-null    object
 4   Region      230 non-null    object
 5   Product     230 non-null    object
 6   Category    230 non-null    object
 7   Sales       230 non-null    int64
 8   Quantity    230 non-null    int64
 9   Discount    230 non-null    float64
 10  Profit      230 non-null    float64
dtypes: float64(2), int64(3), object(6)
memory usage: 19.9+ KB
None
```

```
# Find number of rows and columns
print(df.shape)
```

```
(230, 11)
```

```
# Identify data types of each column
print(df.dtypes)
```

```
Order_ID        int64
Order_Date     object
Customer       object
City           object
Region         object
Product        object
Category       object
Sales           int64
Quantity        int64
Discount      float64
Profit        float64
dtype: object
```

Q2 Sales Filtering Analysis Filter the dataset to display: 1.Orders where Sales > 3500 2.Orders where Region = "East" and Sales > 2500 Explain what insights you observe.

```
#1
df[df['Sales']>3500]
```

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
4	1005	2024-10-15	Rahul	Bangalore	South	Monitor	Electronics	3550	2	0.20	146.30
7	1008	2024-03-18	Kavya	Kolkata	East	Phone	Electronics	4693	7	0.05	918.49
8	1009	2024-10-31	Kavya	Ahmedabad	East	Phone	Electronics	5071	6	0.10	707.55
10	1011	2024-06-30	Neha	Delhi	South	Printer	Office	5977	6	0.00	1578.32
...	...	...	...	...	...	...	...	...	...	...	...
219	1220	2024-03-22	Meera	Bangalore	West	Phone	Electronics	5986	7	0.05	986.70
222	1223	2024-03-04	Meera	Ahmedabad	South	Phone	Electronics	4356	6	0.10	416.87
223	1224	2024-01-07	Dev	Bangalore	East	Tablet	Electronics	4115	4	0.05	870.03
225	1226	2024-06-14	Anita	Ahmedabad	East	Laptop	Electronics	5647	2	0.15	662.00
229	1230	2024-04-13	Meera	Kolkata	North	Monitor	Electronics	5087	8	0.10	416.72

104 rows × 11 columns

```
#2
df[(df["Region"] == "East") & (df["Sales"] > 2500)]
```

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
7	1008	2024-03-18	Kavya	Kolkata	East	Phone	Electronics	4693	7	0.05	918.49
8	1009	2024-10-31	Kavya	Ahmedabad	East	Phone	Electronics	5071	6	0.10	707.55
13	1014	2024-03-18	Anita	Bangalore	East	Mouse	Accessories	3616	4	0.05	463.93
21	1022	2024-01-14	Nitin	Chennai	East	Monitor	Electronics	4827	6	0.05	912.34
27	1028	2024-07-21	Dev	Kolkata	East	Tablet	Electronics	4054	6	0.00	1013.83
33	1034	2024-12-15	Nitin	Ahmedabad	East	Tablet	Electronics	4378	2	0.20	-104.30
34	1035	2024-02-20	Nitin	Ahmedabad	East	Laptop	Electronics	2535	4	0.10	188.15
41	1042	2024-07-15	Rohan	Chennai	East	Laptop	Electronics	3268	8	0.15	295.15
43	1044	2024-09-20	Kavya	Hyderabad	East	Printer	Office	2618	7	0.05	454.64
52	1053	2024-05-24	Anita	Chennai	East	Phone	Electronics	5949	7	0.20	220.44
55	1056	2024-01-03	Sneha	Chennai	East	Headphones	Accessories	2784	4	0.05	474.60
58	1059	2024-12-30	Anita	Delhi	East	Tablet	Electronics	4790	1	0.20	238.53
60	1061	2024-02-05	Nitin	Delhi	East	Monitor	Electronics	4508	5	0.00	972.35
70	1071	2024-05-28	Vikas	Hyderabad	East	Mouse	Accessories	3762	2	0.05	719.68
71	1072	2024-05-20	Rahul	Chennai	East	Printer	Office	3696	2	0.10	600.75
73	1074	2024-10-10	Sara	Bangalore	East	Mouse	Accessories	4004	1	0.15	445.81
74	1075	2024-11-25	Pooja	Ahmedabad	East	Laptop	Electronics	3865	8	0.20	174.96
76	1077	2024-03-02	Neha	Pune	East	Monitor	Electronics	2964	8	0.20	109.33
81	1082	2024-08-16	Isha	Chennai	East	Phone	Electronics	2720	4	0.15	163.13
87	1088	2024-07-29	Pooja	Mumbai	East	Headphones	Accessories	4811	4	0.20	-25.47
92	1093	2024-01-13	Karan	Kolkata	East	Laptop	Electronics	5372	3	0.15	136.64
96	1097	2024-06-07	Isha	Delhi	East	Phone	Electronics	5096	4	0.15	283.66
98	1099	2024-07-26	Rahul	Kolkata	East	Laptop	Electronics	5674	7	0.05	1141.51
108	1109	2024-10-23	Raj	Hyderabad	East	Tablet	Electronics	2509	4	0.05	371.67
109	1110	2024-11-28	Priya	Ahmedabad	East	Printer	Office	2561	4	0.20	57.96
112	1113	2024-11-01	Meera	Bangalore	East	Tablet	Electronics	4179	5	0.00	754.64
113	1114	2024-05-10	Rahul	Bangalore	East	Mouse	Accessories	3285	3	0.00	625.46
116	1117	2024-12-07	Pooja	Kolkata	East	Tablet	Electronics	3758	5	0.15	446.54
128	1129	2024-06-09	Nitin	Delhi	East	Laptop	Electronics	5713	3	0.00	1498.02
139	1140	2024-05-04	Karan	Ahmedabad	East	Monitor	Electronics	3500	8	0.05	467.16
141	1142	2024-01-01	Rahul	Mumbai	East	Headphones	Accessories	5430	8	0.05	741.25
146	1147	2024-05-31	Raj	Delhi	East	Keyboard	Accessories	5709	7	0.20	383.58
152	1153	2024-02-27	Rahul	Mumbai	East	Phone	Electronics	3644	2	0.05	512.25
156	1157	2024-05-27	Neha	Kolkata	East	Laptop	Electronics	4906	4	0.00	1115.89
171	1172	2024-09-22	Neha	Hyderabad	East	Headphones	Accessories	4228	1	0.20	236.15
179	1180	2024-12-13	Neha	Ahmedabad	East	Monitor	Electronics	2690	7	0.10	452.70
182	1183	2024-12-01	Pooja	Chennai	East	Laptop	Electronics	5884	1	0.05	1266.10
185	1186	2024-03-12	Kavya	Ahmedabad	East	Phone	Electronics	5047	2	0.10	564.78
187	1188	2024-07-10	Isha	Kolkata	East	Mouse	Accessories	4333	8	0.10	667.53
193	1194	2024-07-23	Pooja	Delhi	East	Laptop	Electronics	3767	7	0.20	103.23
197	1198	2024-10-16	Rahul	Ahmedabad	East	Tablet	Electronics	3955	1	0.10	284.92
202	1203	2024-05-10	Rahul	Mumbai	East	Phone	Electronics	2901	8	0.15	252.25
207	1208	2024-04-29	Dev	Chennai	East	Monitor	Electronics	3585	7	0.15	287.44
216	1217	2024-08-16	Karan	Chennai	East	Monitor	Electronics	2643	5	0.15	78.53
223	1224	2024-01-07	Dev	Bangalore	East	Tablet	Electronics	4115	4	0.05	870.03
225	1226	2024-06-14	Anita	Ahmedabad	East	Laptop	Electronics	5647	2	0.15	662.00
228	1229	2024-09-04	Priya	Pune	East	Tablet	Electronics	2635	8	0.00	531.76

Q3 Customer Purchase Analysis Identify the top 5 customers based on total sales. Hint: Use GroupBy with aggregation

```
# Group by customer and calculate total sales
sales = df.groupby("Customer")["Sales"].sum()

# Sort by total sales (highest → lowest)
sort_sales = sales.sort_values(ascending=False)

# Display top 5 customers
sort_sales.head(5)
```

Customer	Sales
Nitin	59820
Raj	58416
Rahul	52291
Dev	48500
Isha	46818

dtype: int64

Q4 Regional Sales Performance Calculate the total sales generated by each region. Then identify: Which region generated the highest revenue?

```
# Group by Region and calculate total sales
region_sales = df.groupby("Region")["Sales"].sum()

# Identify region with highest revenue
highest_region = region_sales.idxmax()

print("Total Sales by Region:\n", region_sales)
print("\nRegion with Highest Revenue:", highest_region)
```

```
Total Sales by Region:
 Region
East     232675
North    158210
South    183108
West     170227
Name: Sales, dtype: int64

Region with Highest Revenue: East
```

Q5 Product Performance Analysis Calculate the total quantity sold for each product. Which product appears to be the most popular?

```
# Group by Product and calculate total quantity sold
product_quantity = df.groupby("Product")["Quantity"].sum()

# Identify most popular product
most_popular = product_quantity.idxmax()

print("Total Quantity Sold by Product:\n", product_quantity)
print("\nMost Popular Product:", most_popular)
```

```
Total Quantity Sold by Product:
 Product
Headphones    115
Keyboard      110
Laptop        140
Monitor       156
Mouse         122
Phone         151
Printer       108
Tablet        150
Name: Quantity, dtype: int64

Most Popular Product: Monitor
```

Q6 Multi-Level Grouping Group the dataset by: Region and Category Calculate the total sales for each combination. What insights can you derive from this analysis?

```
region_category_sales = df.groupby(["Region", "Category"])["Sales"].sum()

print(region_category_sales)
```

```
Region  Category
East    Accessories     57646
        Electronics    160907
        Office          14122
North   Accessories     62052
        Electronics     70023
        Office          26135
South   Accessories     80914
        Electronics     86690
        Office          15504
West    Accessories     64811
        Electronics     84128
        Office          21288
Name: Sales, dtype: int64
```

Q7 Pivot Table Creation Create a pivot table showing: Rows → Region Columns → Product Values → Total Sales Which region sells the most laptops?

```
pivot = pd.pivot_table(
    df,
    values="Sales",
    index="Region",
    columns="Product",
    aggfunc="sum",
    fill_value=0
)
(pivot)
```

Product	Headphones	Keyboard	Laptop	Monitor	Mouse	Phone	Printer	Tablet
Region								
East	23899	10845	55568	31299	22902	39667	14122	34373
North	15450	20538	4540	14922	26064	30535	26135	20026
South	24111	23480	23029	19167	33323	18218	15504	26276
West	17159	38499	10826	18178	9153	27850	21288	27274

Next steps: Generate code with `pivot` New interactive sheet

Q8 Profitability Analysis Calculate the average profit for each product category. Which category is most profitable?

```
# Group by Category and calculate average profit
category_profit = df.groupby("Category")["Profit"].mean()

# Identify most profitable category
most_profitable = category_profit.idxmax()

print("Average Profit by Category:\n", category_profit)
print("\nMost Profitable Category:", most_profitable)
```

```
Average Profit by Category:
 Category
Accessories    410.603086
Electronics    396.462114
Office         471.938462
Name: Profit, dtype: float64

Most Profitable Category: Office
```

Q9 Discount Impact Study Analyze the relationship between discount and profit. Tasks: Calculate the average profit for each discount level Identify whether higher discounts lead to lower profit

```
# Group by Discount and calculate average profit
discount_profit = df.groupby("Discount")["Profit"].mean()

print("Average Profit by Discount Level:\n", discount_profit)

# Trend: Higher discounts lead to lower profit
```

```
Average Profit by Discount Level:
 Discount
0.00    751.210588
0.05    531.368039
0.10    346.460513
0.15    237.191400
```

```
0.20     90.026923
Name: Profit, dtype: float64
```

Q10 End-to-End Data Wrangling Task Perform the following workflow: 1.Filter the dataset where Sales > 3000 2.Group data by Region and Product 3.Calculate total sales 4.Reshape the result into a pivot table Explain the business insights obtained from this analysis.

```
# Step 1: Filter rows where Sales > 3000
filtered = df[df["Sales"] > 3000]

# Step 2 & 3: Group by Region and Product, calculate total sales
grouped = filtered.groupby(["Region", "Product"])["Sales"].sum().reset_index()

# Step 4: Reshape into pivot table
pivot = grouped.pivot_table(
    values="Sales",
    index="Region",
    columns="Product",
    aggfunc="sum",
    fill_value=0
)

pivot
```

Product	Headphones	Keyboard	Laptop	Monitor	Mouse	Phone	Printer	Tablet
Region								
East	14469	9642	44096	16420	19000	29500	3696	29229